TECHNICAL GUIDE · OCTOBER 2026

Retirement Optimizer

The objective, how candidate strategies are built, how the staged Monte Carlo ranks them, and four households followed through every step.

Every figure is from backend/benchmarks/walkthrough.py, seed 42, 300 market scenarios, run on 21 October 2026.

This is an open-source planning tool for education and exploration, not personalized financial or tax advice. Results depend on assumptions and may contain errors. Review actual decisions with a CFP® professional and/or CPA.

The objective

```
score = mean over market scenarios of
  PV(heir-adjusted terminal estate) – λ · shortfall_pv – λd · discretionary_shortfall_pv
```

Estate

Final-year balances with traditional dollars counted at (1 – heir tax rate, 25%), discounted to the start year at a fixed **2%**. Taxes enter only through the estate — never subtracted twice.

Penalties

λ = 1: one PV dollar of essential spending unfunded costs one dollar. λd = 0.5 for spending cut above the essential floor. A shortfall never ends a path; the years after count.

Scenarios

300 scenarios, 12% volatility, one independent return per year. Scenario *r* is Random(seed + r), so every strategy sees the same sequences: comparisons are paired.

There is no variance term: a standard deviation would penalize a strategy for occasionally producing a larger estate. The same objective_value scores the steady-return projection (one path) and the Monte Carlo (the mean over paths).

Two decision variables; everything else is input

The conversion schedule

One dollar amount per plan year. Chosen from a finite menu of rules and their stop-year variants (next slides), never free-form. Each year is solved with the full federal, state, Medicare-surcharge, RMD and Social Security engines (transition.simulate_year).

The claim ages

Optional, per person: every eligible age 62–70 (claiming.claim_age_options), chosen jointly with the schedule in joint.run_joint_optimization. Fixed ages make the schedule the whole problem.

Held fixed

Spending (inflated at the plan's inflation), the mean return and one allocation for every account, inflation, horizon, bequest target, withdrawal order, IRMAA policy, bracket cap, conversion window. Not modeled: mortality in the main projection, the earnings test, charitable giving, tax-loss harvesting, capital-loss deductions.

The menu: nine rules, built by search()

Policy id	Label on the page	Amount each allowed year
`none`	No conversions	All-zero schedule, registered first; the counterfactual for every comparison.
`fill_10 … fill_32 _annual`	Fill the N% bracket every year	The conversion that lands taxable ordinary income on that bracket's top — closed form, then bisection on the simulated year when it overshoots.
`level_5pct/10pct/20pct _annual`	Convert up to $X annually	share × starting traditional balance, sized once and passed as a requested maximum; caps, affordability and IRMAA/rate filters still apply.
`<rule>_thru_YYYY`	… through YYYY	The rule through YYYY, zero after — a prefix of the full walk (next slide).

Generated but never requested. `generate_conversions` still sizes `fill_irmaa_tier`, `level_25/50/100` (¼, ½, 1× of balance ÷ 5, the retired greedy window) and `max_pre_rmd`; the search calls it with `only=rule_id`, so none becomes a policy.

Why these three shares. The greedy rollouts were retired in October 2026 after a gate showed that the fill rules plus a level-dollar family land within max($5,000, 0.25% of assets) of greedy on every case. A benchmark result, not a finding that 5/10/20% are optimal.

One walk per rule; variants are prefixes

1 · Walk — `_build_rule_schedule`

For each year of the horizon, on the steady-return projection: regenerate the rule's amount on the evolved state, record it, simulate the year, move on. The window gate, yearly cap, IRMAA policy and marginal-rate limit apply because the same `generate_conversions` is used.

2 · Variants — `_duration_variants`

For n = 1 … last converting year: `schedule[:n] + zeros`. A year's amount depends only on that year's state, which depends only on earlier conversions, so the prefix is exactly "apply the rule through year n". No re-walk; one `simulate_path` per variant.

3 · Dedup — `_dedup_key`

Schedules identical after trailing zeros are trimmed are kept once, under the first-registered name — every `_annual` rule before any variant, so a full-length variant keeps the rule's name.

Household (walkthrough case)	Horizon	Ceiling 8 × H + 1	Policies built
Comfortably funded couple, $2.5M IRA (happy)	30 years	241	97
Small IRA, tight budget, $500k (tight)	30 years	241	43
Very large IRA, $6M (large)	30 years	241	112
Still working, $1M IRA, salaries to 2031 (working)	35 years	281	76

What shrinks a schedule, and what removes one

Situation in a year	Behaviour	Where
Amount exceeds the IRA or `max_conversion_per_year`	Clamped, then re-checked against the $1,000 minimum.	`generate_conversions`
Conversion would leave the year shorter of spending money than converting nothing	Cut back to what the year can spare; spending is funded from start-of-year balances, so converted dollars cannot pay for the same year.	`_affordable_conversion`
Bracket already full, window closed, IRMAA tier or rate limit crossed	$0 that year; the rule is asked again next year.	`generate_conversions`
The rule converts $0 in every year	Skipped: its dedup key is empty, the same as `none`.	`search`
Year 0 reports a shortfall	Rule dropped (`feasible=False`); `none` is kept and scored with its shortfalls.	`_build_rule_schedule`
A later year reports a shortfall	The walk breaks; the schedule is shorter than the horizon. `simulate_path` still runs to the end and prices every unfunded year.	`_build_rule_schedule`

In Monte Carlo, `simulate_path` re-applies the affordability cut in every scenario against that scenario's balances: a conversion is reduced to whatever leaves the year no shorter than converting nothing would. A shortfall the market causes anyway still counts in that scenario's objective. The ledger records what was actually converted.

Fixed claim ages — run_optimization

1–3 · search()

Walk the 8 rules, add prefix variants, dedup, add none. Sorted by path_value.

4 · Steady-return rank

Informational here: every policy goes to stage 1 regardless. It is the screen only in the claim-age search, and the ranking itself when monte_carlo_runs=0.

5 · Stage 1

Every policy on scenarios [0, 100) — runs // 3. Shared sample, paired ranking.

6 · Stage 2

Top 3 by stage-1 mean, plus none (contender) and any forced policy (pinned), extended with scenarios [100, 300).

7 · Rank

Finalists and the contender by score, then success probability to break a tie. A pinned policy cut at stage 1 keeps its stage-1 rank: more scenarios do not promote it. Eliminated policies follow, with their path count.

8 · compare_paired

Winner vs each finalist, scenario by scenario: mean, median, 95% CI, verdict. Then build_report.

Appending, not redrawing. Scenario *r* is the same return sequence at any budget, so stage 2 adds 200 scenarios to the 100 already run. 500 runs would keep scenarios 0–299 byte-identical and add 200 more.

A policy cut at stage 1 cannot win at any budget. It lost on the evidence available; a noisier estimate is not grounds to reinstate it. monte_carlo_paths on every policy makes the cut visible in the exports.

When the two rankings disagree

`path_value` is the objective on one expected-return path. It cannot see sequence risk, so with market simulation on the decision is never taken on it (with `monte_carlo_runs=0` it is the whole ranking) — but in a claim-age search it is the per-pair screen (top 3 + `none`), so where it disagrees with the Monte Carlo matters.

Happy — agreement at the top

Strategy	Steady rank	Stage-1 rank
Fill the 12% bracket every year	1	1
Fill the 12% bracket through 2037	2	2
Fill the 12% bracket through 2036	3	3
Convert up to 5% ($125,000) through 2037	5	7
No conversions	77	80

The same three finalists either way; the disagreement is below the cut.

Large — the cut bites

Strategy	Steady rank	Stage-1 rank
Fill the 22% bracket every year	1	1
Fill the 22% bracket through 2037	2	2
Fill the 12% bracket every year	3	5
Fill the 22% bracket through 2036	4	3
No conversions	100	102

The 12%-bracket family was third on the steady path, fifth on 100 scenarios, and never received the full budget. Only the 22% family was fully scored.

The staged budget, on the happy household

Strategy	Steady rank	Mean, 100 scenarios	Mean, 300 scenarios
Fill the 12% bracket every year	1	$5,973,540	$5,816,505
Fill the 12% bracket through 2037	2	$5,961,674	$5,804,753
Fill the 12% bracket through 2036	3	$5,949,087	$5,792,438
Fill the 12% bracket through 2035	4	$5,935,480	eliminated at stage 1
No conversions (contender)	77	$5,646,947 (80th)	$5,498,744

Averages move, the order does not

The first 100 scenarios happened to be kinder markets; all three finalists dropped about $157,000 on the full sample and kept their order. 300 runs picks a policy; it does not pin a success probability to the percent.

`contenders="none"`

The baseline is extended to the full budget whatever its stage-1 rank **and** ranked with the finalists, so “do nothing” can win after full evaluation and every gap to it rests on 300 shared scenarios.

`pinned` (a forced policy)

A strategy forced from the What-if page is measured on the full budget, so its interval rests on the count the page advertises — but it keeps the rank stage 1 gave it. Measuring it more does not promote it.

Paired verdicts — compare_paired

Label	Rule
Practically equivalent	Checked first: the whole 95% CI of the per-scenario difference lies within ± materiality = max($5,000, 0.25% of investable assets). Evidence of a small difference — even if the interval excludes zero.
Recommendation ahead	Otherwise, the interval excludes zero and the median difference has the mean's sign.
Alternative ahead	The same, in the alternative's favour.
Too close to distinguish	Anything else. Not enough evidence — a different conclusion from equivalence.

Both policies saw identical return sequences, so the differences are paired and about 3.8× tighter than comparing independent means. The comparison also reports how many scenarios each side alone survived, because a few insolvent scenarios can dominate a mean.

Working household

$4,694 lead → "ahead"

Less than the $5,000 threshold, yet *Recommendation ahead*: the interval [−$5,952, −$3,437] excludes zero and reaches past the threshold, so it is not equivalence.

Tight household

$4 and $44 → "equivalent"

Intervals [−$55, $47] and [−$157, $69], both inside ± $5,000: small conversions and none perform alike. The pinned fill_32 trails by $288,738 — larger conversions are clearly worse.

Example 1 — a comfortably funded couple

62/62, MFJ, California; $400,000 taxable (basis $300,000), $2,500,000 traditional, $50,000 Roth; $30,000 pension; Social Security $2,800 / $1,600 at FRA, claimed at 67; $90,000 spending; 30 years from 2026. **97 strategies.**

Rule	Years converting	First – last	Total converted	Stop-year variants kept
Fill the 12% bracket	13	2026 – 2038	$983,746	12
Fill the 22% bracket	16	2026 – 2041	$2,757,621	15
Convert up to 5% ($125,000)	28	2026 – 2053	$3,500,000	27
Convert up to 20% ($500,000)	4	2026 – 2029	$2,000,000	3

Winner · Fill the 12% bracket every year

$103,000 in 2026; $107,000–$123,000 a year through 2030; $48,000–$58,000 from 2031 once Social Security takes bracket room. Score $5,816,505; 100% of scenarios funded.

Finalists vs winner

… through 2037: −$11,752, CI [−$12,882, −$10,623] · … through 2036: −$24,066 · No conversions: −$317,761, CI [−$349,072, −$286,449]. All *Recommendation ahead*; materiality $7,375.

Example 2 — a small IRA and a tight budget

65/63, MFJ, California; $500,000 traditional and nothing else; Social Security $2,000 / $1,200 at FRA, claimed at 67; $60,000 spending; 30 years from 2026. **43 strategies.**

Rule	Years converting	Total converted	First shortfall, steady path	Note
Fill the 10% bracket	14	$203,866	2048	$1,340 in 2026: IRA spending fills the bracket
Fill the 32% bracket	1	$299,850	2039	$299,850 of $500,000 in year one
Convert up to 10% ($50,000)	6	$275,935	2046	$25,935 in 2031: cut to what was left
No conversions	0	$0	2050	Latest of the 43; range 2039–2050

Steady vs stage 1

path_value puts *none* first; the first 100 scenarios put two tiny 10%-bracket variants ahead of it by $64 and $73. All three were fully scored. Dates above are depletion_year on the steady path; every scenario has its own.

Finalists, 300 scenarios

Fill 10% through 2027: −$142,418, 23.7% funded · through 2026: −$4 · *none*: −$44, both *Practically equivalent* (threshold $5,000). Pinned for contrast, *fill_32_annual* (43rd of 43 at stage 1): −$431,156, 0.7% funded, −$288,738 vs the winner.

A negative score means the shortfall penalty exceeds the mean PV estate; it does not mean every scenario ends empty — 23.7% fund every year. Small conversions and none are indistinguishable here; larger conversions are clearly worse. Conversions alone do not resolve the gap.

Example 3 — a very large IRA

62/62, MFJ, California; $500,000 taxable (basis $400,000), $6,000,000 traditional, $100,000 Roth; Social Security $3,500 / $2,000 at FRA, claimed at 70; $80,000 spending; 30 years from 2026. **112 strategies.**

Rule	Years converting	Total converted	Variants kept	Note
Fill the 22% bracket	13	$2,412,625	12	The winning family
Fill the 24% bracket	20	$6,773,281	19	More than the starting IRA: it grows while drained
Convert up to 5% ($300,000)	27	$8,036,772	26	The longest walk
Convert up to 20% ($1,200,000)	4	$4,128,277	3	Empties the IRA in four years

Winner · Fill the 22% bracket every year

$243,600 in 2026; $2,412,625 over 13 years. Score $14,321,468; 100% funded. Through-2037 variant −$30,120 [−$33,246, −$26,994]; no conversions −$817,381.

Two things to notice

No conversions ranks 100th of 112 on the steady path and 102nd at stage 1 — RMDs on $6M make almost any conversion better. Materiality scales to $16,500 (0.25% of $6.6M). And the stage-1 cut removed the 12%-bracket family (previous slide).

Example 4 — still working

58/56, MFJ, California; salaries $150,000 (through 2031) and $60,000 (through 2029); $200,000 taxable (basis $150,000), $1,000,000 traditional; Social Security $3,200 / $1,500 at FRA, claimed at 67; $110,000 spending; 35 years from 2026. **76 strategies.**

Rule	First conversion	Years converting	Why
Fill the 10% bracket	2032 · $68,000	5	The first year with no salary
Fill the 12% bracket	2031 · $4,150	15	The indexed bracket just outgrows the last $150,000 salary; $158,750 in 2032
Fill the 22% bracket	2026 · $33,600	10	The salaries leave room above them in the 22% bracket

Winner · Fill the 10% bracket every year

$0 in 2026 — the decision for this year, not a missing strategy. $287,471 over 2032–2036. Score $810,059; spending funded in 57.7% of scenarios; no conversions –$92,294.

Nothing is dropped for a working household

The low-bracket schedules start late; *no conversions* ranks 21st on both rankings, and the 55 strategies below it all convert during the salary years at 22% and above. Every finalist funds 56–58% of scenarios, so the score separates them.

The claim-age search — joint.py

Example 1 run again with both ages searched and 65/63 entered as the comparison ages. Materiality $7,375.

81
claim-age pairs, 62–70 each, all eligible; search() ran once per pair

420
candidates into stage 1: top 3 by path_value + none per pair; every policy at 65/63 (pinned)

50 → 300
stage-1 scenarios, then the top 5 eligible by stage-1 mean on all 300; none at the winner's ages fully scored

358 s
for the whole search, against about a minute at fixed ages

Ages	Strategy	Score	vs winner	Verdict
70/70	Fill the 12% bracket every year	$5,898,802	—	winner
70/69	Fill the 12% bracket every year	$5,894,935	−$3,867	Practically equivalent
69/70	Fill the 12% bracket every year	$5,891,816	−$6,986	Recommendation ahead
65/63	Your ages, best conversions (same rule)	$5,710,417	−$188,385	Recommendation ahead
65/63	Your ages, no conversions	$5,405,983	−$492,819	Recommendation ahead

The five finalists were the same rule at neighbouring ages (70/70, 70/69, 69/70, 69/69, 70/68); the cut fell at 69/68, about $2,700 behind on the first 50 scenarios. Lifespan checks then re-score the top 3 at horizon ±5 and each death at 78.

What the search cannot find

By construction

- An amount between the menu's steps, or a fill to a point inside a bracket.
- A later start chosen for its own sake: a rule converts from the first year it has room; only a salary or the conversion-window setting delays it.
- A strategy cut at stage 1 (fixed ages) or outside a pair's top 3 by path_value (claim-age search).
- Anything the per-year model does not represent — mortality, the earnings test, charitable giving, capital-loss deductions.

What it does guarantee

- Every strategy is simulated over the whole plan with the full tax engine. Finalists and baselines see all 300 market scenarios; a strategy cut at stage 1 saw the first 100. Any two strategies compared saw the same scenarios.
- Nothing is ranked above *no conversions* without being measured on the same evidence.
- A strategy that runs the money out is scored with its shortfalls, not hidden.
- Every strategy's scenario count is reported, so the screening is visible.
- Property tests on hundreds of random households: a fill never lands above its bracket, every year funds spending unless the accounts are empty, RMDs start at the SECURE 2.0 age.

Reproducing every number

```
cd backend
uv run python -m benchmarks.walkthrough --out /tmp/walkthrough.json
uv run python -m benchmarks.walkthrough --case tight
uv run pytest tests/unit/test_walkthrough_cases.py
```

The walkthrough

Defines the four households in code, runs `search` and `score_policies_adaptive` one stage at a time, and prints every table in the written guide. Eight unit tests guard the properties each example illustrates.

The benchmarks

`greedy_vs_rules` re-runs the gate that retired the greedy search; `joint_screening` measures what the claim-age screen leaves out. A confirmation seed must differ from 42 by at least the path count.

Verifying a year by hand

The Excel Tax Audit sheet carries the 1040 derivation per year. Identity, to within $1: cash income + RMD + spending withdrawals − total tax = required spending + surplus saved − shortfall. Conversions are not spending withdrawals.

THE ANNUAL CADENCE

Review this year’s action. Re-run next year with real balances and new tables.

Tax data lives in cited files under backend/src/data/tax_rules/ with an annual update checklist; past the latest published year, federal bracket thresholds and deductions, IRMAA tiers and California’s brackets are projected at the plan’s inflation with each provision’s statutory rounding; other states stay at their latest published schedule, and thresholds fixed in law (NIIT, Social Security taxation, the IRMAA top tier through 2027) do not move. The first year’s figure is the one to act on — with a CFP® professional or CPA — and the plan rolls forward with you.